

Installing Helm

MANAGING KUBERNETES WITH HELM

Installing Helm

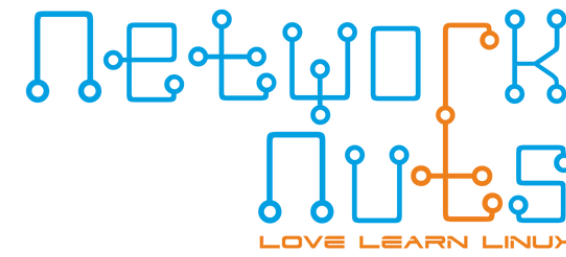
Requirements

You should have access to running Kubernetes cluster. Kubectl is need on your workstation, if you are accessing kubernetes cluster from remote machine.

Installing on Ubuntu Machine using apt repository.

```
curl https://baltocdn.com/helm/signing.asc | gpg --dearmor | sudo tee /usr/share/keyrings/helm.gpg > /dev/null
sudo apt-get install apt-transport-https --yes
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/helm.gpg]
https://baltocdn.com/helm/stable/debian/ all main" | sudo tee /etc/apt/sources.list.d/helm-stable-debian.list
sudo apt-get update
sudo apt-get install helm
```

Configuring Helm



Adding Upstream Repositories

Helm, being the Kubernetes package manager, can connect to various chart repositories to install Kubernetes applications. Helm provides the repo subcommand to allow users to manage configured chart repositories. This subcommand contains additional subcommands that can be used to perform actions against specified repositories. Here are the five repo subcommands:

- add: To add a chart repository
- list: To list chart repositories
- remove: To remove a chart repository
- update: To update information on available charts locally from chart repositories
- index: To generate an index file, given a directory containing packaged charts

Configuring Helm

Adding Plugins

Plugins are add-on capabilities that can be used to provide additional features to Helm. Helm community maintains a variety of different plugins that can be used to enhance Helm's capabilities outside of the core code base. Helm provides a **plugin** subcommand for managing plugins, which contain further subcommands, as described in the following table.

Plugin Subcommand	Description	Usage
<code>install</code>	Installs one or more Helm plugins	<code>helm plugin install \$URL</code>
<code>list</code>	Lists installed Helm plugins	<code>helm plugin list</code>
<code>uninstall</code>	Uninstalls one or more Helm plugins	<code>helm plugin uninstall \$PLUGIN</code>
<code>update</code>	Updates one or more Helm plugins	<code>helm plugin update \$PLUGIN</code>

Configuring Helm

Adding Plugins

Plugins can provide a variety of different productivity enhancements. The following are several examples of upstream plugins:

- Helm Diff: Performs a diff between a deployed release and proposed Helm upgrade
- Helm Secrets: Used to help conceal secrets from Helm charts
- Helm Monitor: Used to monitor a release and perform a rollback if certain events occur
- Helm Unitest: Used to perform unit testing on a Helm chart

Configuring Helm

Environment Variables

Helm relies on the existence of environment variables to configure some of the low-level options. There are many variables you can configure, each of which can be seen in the helm help output. A few environment variables are used for storing Helm metadata:

- HELM_CACHE_HOME or XDG_CACHE_HOME: Sets an alternative location for storing cached files
- HELM_CONFIG_HOME or XDG_CONFIG_HOME: Sets an alternative location for storing Helm configuration
- HELM_DATA_HOME or XDG_DATA_HOME: Sets an alternative location for storing Helm data

Helm adheres to the XDG Base Directory Specification, which is designed to provide a standardized way of defining where different files are located on an operating system's filesystem. Based on the XDG specification, Helm automatically creates three different default directories on each operating system as required:

Operating System	Cache Path	Configuration Path	Data Path
Windows	%TEMP%\helm	%APPDATA%\helm	%APPDATA%\helm
macOS	\$HOME/Library/Caches/helm	\$HOME/Library/Preferences/helm	\$HOME/Library/helm
Linux	\$HOME/.cache/helm	\$HOME/.config/helm	\$HOME/.local/share/helm

Configuring Helm

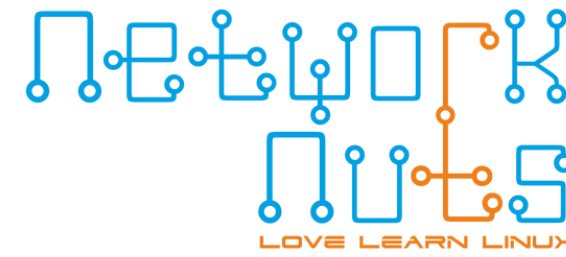
Environment Variables

Helm uses the **cache path** to store charts that are downloaded from upstream chart repositories. Installed charts are cached to the local machine to enable faster installation of the chart the next time it is referenced. The cache path also includes YAML files that are used to index the available Helm charts from each configured repository. These index files are updated when users run the `helm repo update` command.

The **configuration path** is used to save repository information, such as the URL and credentials for authentication, if required. When a chart is installed but is not located in the local cache yet, Helm uses the configuration path to look up the URL of the chart repository. The chart is then downloaded from this URL.

The **data path** is used to store plugins. When a plugin is installed using the `helm plugin install` command, the plugin itself is stored in this location.

Configuring Helm



Tab Completion

Tab completion can be enabled by running one of the following command.

```
# source <(helm completion bash)
```